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Comparison of executive functions in people with high and low resilience

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A R T I C L E I N F O

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A B S T R A C T

Background: The purpose of the present study was to compare executive functions in people with high and low resilience.
Methods: Based on the results of a structured clinical interview and Connor- Davidson resilience Scale (CD-RISC), a total of 140 adults were assigned to high resilience (n = 70) and low resilience (n = 70) groups. Working memory, inhibition and cognitive flexibility were assessed by N-Back, Go/No-Go and Wisconsin Card Sorting task, respectively.
Result: The results showed that individuals with high resilience compared to individuals with low resilience scored significantly higher on inhibition and cognitive flexibility, but there were no significant differences on working memory.
Conclusion: The current findings suggest a role of high-level cognitive processes in resilience, which in turn may contribute to mental health. Also, the results of this research may be important in the design of therapeutic interventions based on executive functions.

Resilience is the process of adapting well in the face of adversity, trauma, tragedy, threats, or significant sources of stress, or bouncing back from difficult experiences (MacLeod et al., 2016). Resilience involves interaction between individuals' personality characteristics and their living environment (Xing, 2020). Mental health specialists have recently embraced resilience as a means of promoting and improving mental health (Rudwan and Alhashimia, 2018). If resilience is not sufficiently effective in the face of adversity, it can lead to the psychological disorders such depression and anxiety, (Afek et al., 2021). Resilience, in comparison to external protective factors, is relatively stable and can predict an individual's mental health (Rutten et al., 2013). Resilience also includes the effective utilization of cognitive skills to cope with stress and the ability to find the appropriate resources for flexible problem solving (Avci et al., 2013). An accurate understanding of the cognitive mechanisms relating resilience will contribute to our knowledge about factors that relate to this important characteristic (Xing, 2020). Resilience utilizes cognitive processes to overcome adversity (Xing, 2020). Empirical evidence substantiates the assertion that executive functions, which encompass a set of cognitive skills that are positively related to resilience (Bemath et al., 2020).

Executive functions encompass the set of higher-order cognitive abilities that are necessary for purposeful behavior to achieve goals, enabling people to plan ahead and accomplish logical outcomes over the short and long term (Cristofori et al., 2019). Cognitive, emotional, and behavioral action are guided, directed, and managed by executive functions (Cristofori et al., 2019). Three core components of executive function are working memory, inhibition and cognitive flexibility (Diamond, 2013; Xing, 2020; de Maat et al., 2022). Studies of adolescents and young adults have demonstrated a significant positive association between executive functions and resilience (McKee, 2017; Xing, 2020; de Maat et al., 2022).

Working memory, conceptualized by Baddeley (2010), is a multi-component capability with executive control skills that regulates the attention allocation and manipulation of language-based and visuospatial information in immediate memory (Baddeley, 2010). It also involves actively managing and storing phonological, linguistic, and visuospatial information (Baddeley, 2010). Effective performance of executive functions depends on working memory, and it also plays a beneficial role in learning, academic achievement, and interpersonal skills (Bemath et al., 2020). Theoretically, working memory can facilitate resilience by enabling individuals to plan, process, and make appropriate decisions for behavior guidance and emotion regulation, as well as organize verbal or nonverbal information relevant to adverse situations (Bemath et al., 2020). Wingo et al. (2010) found that resilient adults had better nonverbal working memory compared to participants with low resilience (Wingo et al., 2010). In a study, Avci et al. (2013)

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also found a significant relationship between resilience and working memory (Avci et al., 2013).

Inhibitory control involves the ability to control one's attention, behavior, thoughts, and/or emotions. It helps the person override a strong internal or external temptation, and instead do what's more appropriate or needed in the current situation (Diamond, 2013). Inhibition determines an individual's behavioral patterns and thus supports their daily adaptive functioning (Nahum et al., 2023). Intact inhibitory control may contribute to psychological resilience and the ability to cope with adversity, while impaired inhibitory control is a potential risk factor for the onset or the aggravation of mental disorders and mental distress (Afek et al., 2021). The study conducted by Afek et al. (2021) identified a positive relationship between resilience and non-emotional inhibition (Afek et al., 2021). Nahum et al. (2023) found that better inhibitory control and mood were associated with high resilience (Nahum et al., 2023).

Cognitive flexibility is a third executive function that enables the ability to adapt to environmental change and generate new ideas that drive innovation and promote growth and discovery (Cristofori et al., 2019). Resilience is one of the variables that positively correlates with cognitive flexibility (Bemath et al., 2020). The mediating role of life goals in the relationship between cognitive flexibility and resilience in high school students was found by Güleç (2020). Individuals with cognitive flexibility have high resilience and a significant relationship between resilience and cognitive flexibility (Dergisi, 2020). A study by Geyik-Koç et al. (2020) revealed a significant relationship between students' levels of cognitive flexibility and psychological resilience.

Studies of adolescents and young adults have demonstrated a significant positive association between executive functions and resilience (McKee, 2017; Xing, 2020; de Maat et al., 2022). Previous studies have only investigated the relationship between resilience and executive functions. Most of the previous studies did not classify people with high and low resilience, or did not use a diagnostic interview to classify them, so the relationship between the variables may have been caused by other variables that were not controlled. Thus far, there has been no assessment of executive function comparing people with high and low resilience. Such a comparison can facilitate comprehension of the role of high-level cognitive processes in resilience, serving as an influential indicator of mental health. Moreover, the results of this study could help the researchers and clinicians in the design of executive function-focused therapeutic interventions. Thus, comparing the components of executive functions in people with high and low resilience was the purpose of the current study.

1. Method

1.1. Participants

The participants were 140 adults (70 with high resilience and 70 with low resilience) aged 18–40 (M = 26.21, SD = 6.53) who were residents of Sanandaj city. They were invited to participate in the research through an announcement. The participants were evaluated in the university's psychology laboratory. The participants were chosen via purposive sampling. The sample size was calculated using the PASS software, with an alpha level of .05 and a test's power of 85% and medium effect size of .3. 50.7 percent of participants were women and 77.1 percent were single. Of the full sample, 47.1 percent had a high school and diploma, 36.4 percent had Associate's or Bachelor's degrees, and 17.9 percent had Master's or Doctorate degrees. Table 1 illustrates a summary of the descriptive distribution of demographic variables among the participants.

Table 1
Demographic data across conditions.

Demographic	High resilience $n = 70$	Low resilience $n = 70$
Gender (male), n (%)	37 (26.4%)	32 (22.9%)
Marital status (single), n (%)	55 (39.3%)	53 (37.9%)
Education		
high school and below	10 (7.1%)	11 (7.9%)
Diploma and Associate's Degrees	27 (19.3%)	30 (21.4%)
Bachelor's Degrees and above	33 (23.6%)	29 (20.7%)

1.2. Measures

1.2.1. Diagnostic screening measures

The Structured Clinical Interview for Personality Disorders (SCID-5-PD) is a semi-structured diagnostic interview introduced in 2016 by First et al. (First et al., 2016; First et al., 2016). The SCID-5-PD is used to assess 10 DSM-5 personality disorders. This interview consists of a predetermined set of questions that are presented in a specific order (Gharraee et al., 2022). In a study conducted by Maffei et al. (2011) the reliability coefficients between raters ranged from .48 to .98 for classified diagnosis (Cohen's K). Furthermore, the internal consistency ranging from (.94 -.71) was satisfactory (Somma et al., 2017). Another study has demonstrated that, SCID-5-PD has been shown to have excellent interrater reliability in a European sample (Paap et al., 2022). In a study conducted by Gharraee et al., (2022), they examined the ability of the translated version of SCID-5-PD in diagnosing personality disorders. The sensitivity of all diagnoses except avoidant and dependent personality disorders exceeded .8. However, the sensitivity of avoidant and dependent personality disorders was .66.

The Structured Clinical Interview for DSM-5- Clinician Version (SCID-5-CV) was introduced in 2016 by First et al. (2016). It covers diagnoses most frequently seen in clinical settings, as well as 17 additional DSM-5 disorders (First et al., 2016; Osório et al., 2019). This interview assesses the major psychiatric disorders such as (First et al., 2016) as depression and bipolar disorders, schizophrenia spectrum and other psychotic disorders, substance use disorders (SUD), anxiety disorders (panic disorder, agoraphobia, social anxiety disorder (SAD), generalized anxiety disorder (GAD), obsessive-compulsive disorder, post-traumatic stress disorder (PTSD), attention deficit/hyperactivity disorder (ADHD), and adjustment (First et al., 2016). Osório et al. (2019) conducted a study to assess the validity and reliability of SCID-V-CV (Osório et al., 2019). The diagnostic sensitivity/specificity exceeded .7, whereas the positive agreement between the clinical diagnosis and the interview ranged from 73% to 97%. Additionally, the Kappa coefficient exceeded .7. In Iranian samples, Shabani et al. (2021) conducted a study to assess the psychometric properties of SCID-V-C. According to the psychiatrist's diagnosis, the kappa value exceeded .8 for most of the diagnoses, except for anxiety disorders. This implies that the instrument possesses a desirable quality for precisely detecting disorders (Shabani et al., 2021).

The Connor-Davidson Resilience Scale (CD-RICS) is an effective instrument used in both psychiatric and non-psychiatric adults that its purpose assesses resilience indifferent individuals (Connor and Davidson, 2003). This instrument consists of 25 items that assess how the subject has felt over the past month (Connor and Davidson, 2003). Items were rated on a 5-point Likert scale: (0 = not true at all), to 4 = true nearly all of the time). The scores range from 0 to 100. This questionnaire has 5 subscales (Connor and Davidson, 2003). The reliability and validity of the subscales are acceptable. Cronbach's Alpha value of .89 is reported (Burns and Anstey, 2010). In Iranian version of CD-RISC, the Cronbach's alpha coefficient of .89 was obtained (Samani et al., 2007).

1.2.2. Outcome measures

N-Back task was first introduced by Wayne Kirchner in 1958 and assesses an individual's working memory capacity (Sawami et al., 2017). It contains a variety of stimuli such as number and shape. The N-Back task requires participants to decide if each stimulus in a sequence corresponds to the stimulus that occurred n times prior (Harrison, 2017). If the stimulus presented to the participant is the same as the stimulus shown earlier in the series, the participant responds. For instance, if it was 1-Back, the participant responds if the stimulus is the same as the previous stimulus, and if it was 2-Back or 3-Back, the participant responds if the is the same as the stimulus 2 or 3 items back in the sequence, respectively (Harrison, 2017). Participants are required to recall the stimulus associated with to the current level N, prevent interference from other irrelevant stimuli, and constantly update the relevant stimulus (Harrison, 2017). Validity coefficients of .54–.84 has been reported and reliability of this task has been reported highly accepted (Kane et al., 2007).

Go/No-Go task is a widely used test of inhibitory control. In this task one stimulus requires a response (Go stimulus), while the other stimulus does not require a response (No -Go stimulus). The Go/No-Go task involves the presentation of two-colored rectangles, one green and one white. The subject is instructed to respond to the green rectangle in any direction (Go) and to refrain from responding to the white rectangle (No -Go). A total of 40 stimuli are presented, with each stimulus being displayed for a duration of .2–3 s. The interval between two stimulus presentations ranges from 1 to 5 s. Go stimuli account for 70% of all stimuli, indicating a subject's bias towards the Go-response (Georgiou and Essau, 2011). In Iranian samples the reliability coefficients were reported to be .87 (Mirdoraghi et al., 2012).

Wisconsin Card Sorting task (WCST) assesses cognitive flexibility (Diamond, 2013). This task assesses an individual's capacity for changing cognitive strategies in response to environmental feedback. WCT contains 64 cards, with one to four symbols in the shapes of triangle, star, plus, and circle. These symbols are presented in four different colors: red, green, yellow, and blue. Importantly, each card is unique and no two cards are similar. There are four primary and constant cards which have shown on the top of the screen: one red triangle, two green stars, three yellow pluses, and four blue circles. The participants are required to deduce the underlying pattern or rule based on the correct or incorrect feedback they receive after each response. The three sorting rules that the participant must discover are shape, color and number. After the participants give a 10 in a row without any errors, the target pattern is changed and the participant must discover the next correct sorting rule based on the feedback given to subsequent responses (Grant, 1993). The interrater reliability coefficient of this instrument has been reported as .83 (Diamond, 2013) and the validity criteria of this instrument, is reported 86% (Anderson, 2002). In Iranian samples the internal consistency coefficient (Cronbach's Alpha) is reported .7, and the split-half reliability coefficient is estimated to be .8 (Shahgholian et al., 2012).

1.2.3. Procedure

The participants were invited to participate in the study via an announcement. To evaluate inclusion and exclusion criteria, participants were screened using SCID-5-PD and SCID-5-CV. The inclusion criteria for this study were age range 18–40 years, the ability to read and write, not having the physical health and not taking any medications that could affect performance, not having any psychological disorders, no intake psychiatric medications, the avoidance of drug and substance abuse, and not undergoing any therapeutic interventions concurrent with the research. Exclusion criteria were significant psychological changes that occurred throughout the study, a lack of informed consent, and unwillingness to continue in the research.

Among 200 participants in the evaluation sessions, 140 people were qualified and assigned to groups: the group with high resilience (n = 70) and the group with low resilience (n = 70). Table 2 illustrates a summary of the information of individuals who did not qualify to participate in the study. There are various Questionnaires to measure resilience such as the Connor- Davidson Resilience Scale (CD-RISC), the Resilience Assessment Questionnaire (RAQ), Nicholson McBride Resilience questionnaire (NMRQ) and Brief Resilience Scale (BRS). Among them, the CD-RISC was chosen because its validity and reliability have been reported in the population of normal individuals and its content was appropriate for the research purpose. So, individuals were categorized into groups using the CD-RISC. Ebrahimi et al. (2012) reported the mean score of resilience questionnaire among a population of normal individuals as 58.1, with a standard deviation of 7.4. Consequently, those who obtained scores above 66 (M + SD) were categorized into the high resilience group, while those with scores below 50 (M- SD) were placed in the low resilience group (Ebrahimi et al., 2012). The results of individuals who obtained CD-RISC score of between 50 and 66 were not included in the analysis, but they were tested and the results were provided to them. The participants completed the Wisconsin Card Sorting Test (WCST), the Go/No-Go, and the 1-Back simultaneously in a session that lasted around 1 h. A 10-min break was given after each task. All these softwares were implemented in a laboratory with a 22-inch monitor. The instructions of the tasks were explained to the participants and the buttons they needed to answer the tasks were marked on the keyboard. Then the tasks were performed according to their instructions.

Table 2
The information of individuals who did not qualify to participate in the study.

Individuals' characteristics	*n*
Psychological disorders	20
Drug and substance abuse	8
Serious suicidal thoughts	4
Physical illness	7
Taking medications	8
Unwillingness to continue in the research	5
Receiving therapeutic interventions	8

2. Results

2.1. Statistical analysis

Independent *t*-test and chi-square test were administered to consider differences in demographic variables between groups. No significant differences were found between groups in age ($t_{(2,\ 40)} = .73$, $p = 0.49$), gender ($\chi 2 = .71$, $p = 0.39$), marital status ($\chi 2 = .16$, $p = 0.68$), job status ($\chi 2 = 2.31$, $p = 0.12$) and education ($\chi 2 = .46$, $p = 0.79$). This indicates that none of these demographic variables justify membership in the research groups.

To compare the groups (high and low resilience) in executive functions, a multivariate analysis of variance (MANOVA) was conducted. Resilience was the independent variable, and working memory, inhibition, and cognitive flexibility were the dependent variables. Reported *p* values are two-sided. All analyses were performed using IBM SPSS 25 software. All assumptions were checked and met, and there were no missing data.

Kolmogorov-Smirnov's normality test was met for inhibition ($D_{(140)} = .158$, $p > 0.05$), working memory ($D_{(140)} = .160$, $p > 0.05$) and cognitive flexibility ($D_{(140)} = .185$, $p > 0.05$). Before using multivariate analysis of variance, BOX'S M and Levene's tests were used to check the assumptions of the MANOVA. Based on the results of the BOX'S M test, the assumption of the homogeneity of the variance/covariance matrices was met (BOX'S M = 11.814; $F = 1.922$; $p = 0.073$). Also, according to the results of Levene's test, the assumptions of the equality of error variance was met for working memory ($F_{(1,\ 138)} = .008$, $p = 0.928$), inhibition ($F_{(1,\ 138)} = .996$, $p = 0.32$) and cognitive flexibility ($F_{(1,138)} = 2.753$, $p = 0.099$).

2.2. Main analysis

The results of MANOVA showed significant differences between groups ($F_{(3,136)} = 3.012$, $p = 0.032$; Wilks's $\Lambda = .938$, $\eta 2 = .062$). The means (*M*) and standard deviations (*SD*) of groups (high and low resilience) and the univariate F-test for the dependent variables are presented in Table 3. The results showed that there was significant difference between groups in inhibition ($F_{(1,138)} = 4.077$, $p = 0.045$, $\eta 2 = .029$) and cognitive flexibility ($F_{(1,138)} = 4.798$, $p = 0.03$, $\eta 2 = .034$) but there was no significant difference in working memory ($F_{(1,138)} = .251$, $p = 0.617$, $\eta 2 = .002$). The highest effect size was obtained for inhibition.

3. Discussion

The findings of the current study shown a significant difference in executive functions (inhibition, cognitive function) between individuals with higher levels of resilience and lower levels of resilience. The findings of this study are consistent with the research conducted by Parsons et al. (2016), McKee Jackson (2017), Lina Xing (2020), de Matt et al. (2022), Nahum et al. (2023), Afek et al. (2021), Burton et al. (2010), Taghizadeh and Farmani (2014), Güleç (2020), and Geyik-Koç et al. (2020). However, it contradicts the findings of Wingo et al. (2010), Avci et al. (2013), and Bemath et al. (2020).

The result showed that people with greater high resilience scored significantly higher on inhibition compared to people with low resilience. The finding is consistent with the results of previous studies like Schafer et al.'s (2015), Parsons et al. (2016), Davidovich et al. (2016), Afek et al. (2021), and Nahum et al. (2023). Inhibition reflects an individual's behavioral pattern that plays a crucial role in achieving effective performance in many daily life situations. Without inhibitory control we would be at the risk of impulsivity, and interference from our conditioned responses and/or environmental stimuli that distract us from the task at hand. Thus, inhibitory control makes it possible for us to change and choose how we react and behave rather than being unthinking creatures of habit (Diamond, 2013). Inhibition deficits make it more difficult to control negative thoughts that make one vulnerable to mental distress (Afek et al., 2021). Thus, inhibitory control may contribute to psychological resilience and the ability to cope with challenges, while impaired inhibitory control is a potential risk factor for the onset or the aggravation of mental disorders. Furthermore, stronger inhibitory control may reduce emotional reactivity to negative stimuli and support the use of adaptive cognitive reappraisal strategies (Nahum et al., 2023).

Another study finding was that people with high resilience scored significantly higher on cognitive flexibility compared to people with low resilience. This finding is consistent with the results of previous studies conducted by Burton et al. (2010), Taghizadeh and Farmani, (2014), Daniel Busso (2014), Parsons et al. (2016), Davidovich et al. (2016), Güleç (2020), and Geyik-Koç et al. (2020). Cognitive flexibility is important to react appropriately when faced with changing conditions or unexpected events. Cognitive flexibility allows us to adapt to the situation by improving level of resilience, knowing that every situation has options or alternatives (Canas et al., 2003). Individuals with cognitive flexibility can replace compulsive and maladaptive thoughts with those that are appropriate and adaptive. Also, these people consider difficult situations to be controllable (Gülm and Dağ, 2012). Thus, cognitive flexibility may be a construct related to psychological resilience and lead to more adaptive coping against challenging events. Based on the information provided, it can be inferred that individuals with higher cognitive flexibility are more resilient, enabling them to accept stressful situations and reconstruct their thinking framework in a positive manner. Conversely, limited cognitive flexibility may contribute to more vulnerability to stress and lower resilience.

The last finding of our study showed that there is no significant difference between people with high and low resilience on working memory scores. This finding is inconsistent with the previous studies conducted by Wingo et al. (2010), Avci et al. (2013), and Bemath et al. (2020). In the current study employed the 1-Back was administered to compare the individuals' working memory. It may be that individuals with different levels of resilience do not exhibit significant differences on simpler memory tests, but differences may become apparent when memory tasks get more difficult. However, they may have difficulties when they are required to engage in tasks that demand memory use for extended periods of time. The N-Back task used in the current study may not have utilized of all working memory components, and this may be another reason for the lack of group difference in performance on working memory.

This study had several strengths. According to our review of the literature, this was the first study that compared the executive functions on individuals grouped by high and low resilience. The implementation of the structured diagnostic interview (SCID) enabled an accurate evaluation of inclusion and exclusion criteria, as well as determining membership in the study groups. Additionally, a notable advantage was the utilization of software to provide more precise assessment of executive functions. Self-report measures have questionable accuracy in evaluating cognitive abilities, as individuals may only report their personal perspectives of their own functioning. Despite these strengths, this study has some limitations. The present study was carried out on normal individuals without any psychological disorders. Future study should focus on clinical samples to examine the association between resilience and executive functions in individuals with psychological disorders. This study focused solely on examining three core components of executive functions in people. Subsequent studies can explore other components of executive functions within these groups (high and low resilience). Regarding working memory, the 1-back task was used, and it may not be accurate to check the difference between groups. We suggest using more challenging levels such as 2-back and 3-back tasks in future researches. Various questionnaires have been designed to measure resilience. Each of them can be used for specific individuals or groups. For example, Walsh family resilience questionnaire is used to assess family resilience processes (Walsh, 2016). It is recommended to use a questionnaire that is accurate enough to measure resilience according to the target group in future researches. Judgmental sampling increases the relevance of the sample to the population of interest, but may not be representative of the entire population, leading to limited generalizability of the findings (Perla and Provost, 2012). It is recommended to use other random sampling methods in future researches. To examine the executive functions, we evaluated only three main domains including working memory, cognitive flexibility and inhibitory control. It is suggested to evaluate a wider range of variables in future researches.

4. Conclusion

Executive functions are essential for using appropriate problem-solving, adjustment and adaptive strategies when facing new and

Table 3
Descriptive statistics and univariate F-test for dependent variables (Inhibition, cognitive flexibility and working memory).

	High resilience Means (*SD*)	Low resilience Means (*SD*)	*S.S*[a]	*M.S*[b]	$F_{(1,138)}$	*P*	Effect size[c]
Inhibition	38.49 (1.29)	38 (1.54)	8.257	8.257	4.077	.045	.029
Cognitive Flexibility	2.86 (2.96)	4.1 (3.70)	54.064	54.064	4.798	.030	.034
Working Memory	106.26 (12.68)	105.23 (11.60)	37.029	37.029	.251	.617	.002

[a] Sum of Squares.
[b] Mean Square.
[c] Effect sizes reported as partial Eta squared.

stressful situations and can lead to better management of stressful events. Executive functions can facilitate adaptive reactions to the environment that may avoid later stresses or adversities. In the present study, the aim was to highlight the relationship between some executive functions and resilience. Investigating this relationship may contribute to the literature on resilience and executive functions as influential components in mental health. It is important to adopt measures for better performance of executive functions and increase resilience and strengthen mental health in individuals. The current study will lead to the development of rehabilitation and mental health techniques that will help individuals to perform better in work, education and social relationship. This research can help experts and researchers to use or design therapeutic interventions focusing on executive functions and resilience.

Informed consent

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. Informed consent was obtained from participants. All individuals received written information about the research and were asked to participate if interested. Participants were assured that the information would be collected for research purposes only and that their responses would remain confidential. Therefore, participants' names were not recorded. Additionally, they were reminded that they could become aware of the research findings and their results if desired.

Animal rights

No animals were involved in the study.

CRediT authorship contribution statement

Asra Shariati: Writing – original draft, Data curation. **Farzad Nasiri:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Fateh Rahmani:** Writing – review & editing, Conceptualization.

Declaration of competing interest

Asra Shariati, Farzad Nasiri and Fateh Rahmani declare that they have no conflict of interest.

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